

Nasdaq Closing Auction Prediction

Background

The Nasdaq closing auction takes place every day from 3:50pm to 4:00pm. Participants can submit auction orders that are filled at 4pm at the official closing price. The official closing price is determined by Nasdaq based on the submitted auction orders. Between 3:50pm and 4, Nasdaq disseminates auction order information in the form of imbalances. The imbalances contain the following data points

(<https://www.nasdaqtrader.com/content/productservices/Trading/ClosingCrossfaq.pdf>):

- **paired_shares**: The number of on-close shares that Nasdaq is able to pair off at the current reference price.
- **side**: Indicates the direction of the imbalance
- **shares** (Imbalance Quantity): The number of closing order shares that would not be matched at the current reference price.
- **ref_price**: The price within the BBO at which paired shares are maximized
- **near_price**: The closing cross price that will maximize the number of shares matched based on closing auction orders and the continuous limit order book.
- **far_price**: The closing cross price that will maximize the number of shares matched based on closing auction orders only.

The first four data points are disseminated every 10 seconds between 3:50 and 3:55, and every second from 3:55 until 4pm. The last two data points are only disseminated between 3:55 and 4pm.

The goal of this task is to build a model that can predict closing prices, with the intention of entering positions after 3:50pm and exiting them at 4pm with auction orders. You may assume that all auction orders are guaranteed to be filled, even those sent after 3:55pm. The task will involve cleaning the data, analyzing the data, and building a model.

Note: Execution after 3:50pm with the intention of closing positions at 4pm is extremely difficult. There are many basic features which will demonstrate predictive power but will be hard or impossible to monetize. The goal of this task is to demonstrate statistical analysis and modeling skills, not to create monetizable strategies, so assume all execution is free and easy.

Dataset

The dataset (<https://drive.google.com/file/d/1-LfKpi90g5P1NfQvHGZ-5iHT39zmnIA/view>) includes pickled dataframes of Nasdaq market data. Each row represents an imbalance message from the exchange, and contains the imbalance values as well as a few other datapoints:

- **bid**: highest bid price in limit order book
- **ask**: lowest ask price in limit order book
- **bid_qty**: number of shares posted at the highest bid price
- **ask_qty**: number of shares posted at the lowest ask price
- **adv**: average daily traded shares over the last 20 trading days, in thousands of shares
- **cross**: official closing cross price
- **open**: the price that the stock opened at today

Use **pandas** to explore the dataset, determine what you will use as the dependent variable, and perform any necessary cleaning on the dataset.

Analysis

Determine what information is helpful in predicting closing prices. Construct features that may assist in predicting closing prices. Please include code and any plots you use for analysis in your final submission.

Modeling

Construct and evaluate a model to predict closing prices. Include a reproducible Jupyter notebook or Python script in your submission.